WEEK-6

AIM : Implementation of Paging Technique of Memory Management

ALGORITHM:

1. Start.
2. Read page size and number of pages.
3. Read the page table (frame numbers).
4. Read the logical address.
5. Calculate page number = Logical Address / Page Size.
6. Calculate offset = Logical Address % Page Size.
7. If page number ≥ number of pages, display invalid address and stop.
8. Get the frame number from the page table.
9. Calculate physical address = (Frame Number × Page Size) + Offset.
10. Display the address details.
11. Stop.

PROGRAM:

```
#include <stdio.h>
int main() {
    int pageSize, numPages;
    int pageTable[100];
    int logicalAddress;
    int pageNumber, offset, frameNumber, physicalAddress;
    printf("Enter page size: ");
    scanf("%d", &pageSize);
    printf("Enter number of pages: ");
    scanf("%d", &numPages);
    printf("\nEnter the page table:\n");
    printf("(Enter frame number for each page)\n");
    for (int i = 0; i < numPages; i++) {
        printf("Page %d -> Frame: ", i);
```

```
    scanf("%d", &pageTable[i]);   }
    printf("\nEnter logical address: ");
    scanf("%d", &logicalAddress);

    // Calculate page number and offset
    pageNumber = logicalAddress / pageSize;
    offset = logicalAddress % pageSize;

    if (pageNumber >= numPages) {
        printf("Invalid logical address!\n");
        return 1;
    }

    // Get frame number from page table
    frameNumber = pageTable[pageNumber];

    // Calculate physical address
    physicalAddress = (frameNumber * pageSize) + offset;

    printf("\n--- Address Translation ---\n");
    printf("Logical Address  : %d\n", logicalAddress);
    printf("Page Number      : %d\n", pageNumber);
    printf("Offset           : %d\n", offset);
    printf("Frame Number     : %d\n", frameNumber);
    printf("Physical Address : %d\n", physicalAddress);

    return 0;
}
```

VNR VJIET

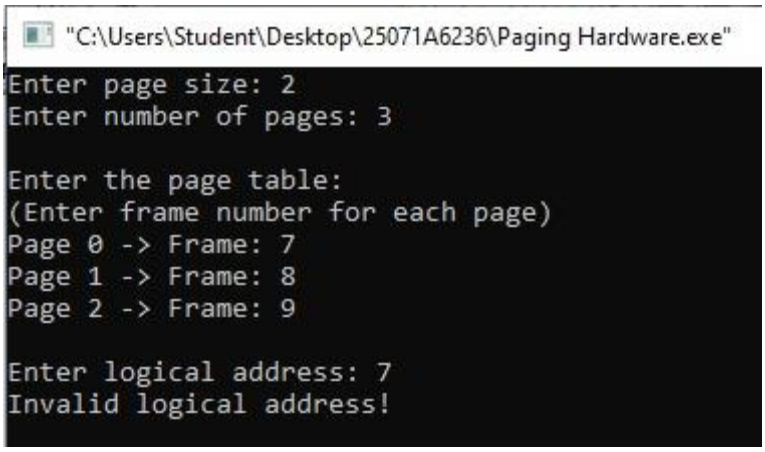
Name of the Laboratory:

Name of the Experiment:

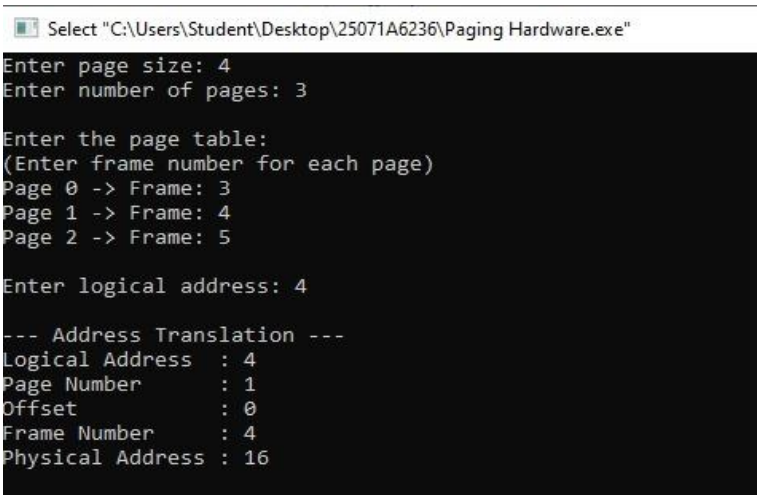
Experiment No: **Date:**

OUTPUT:

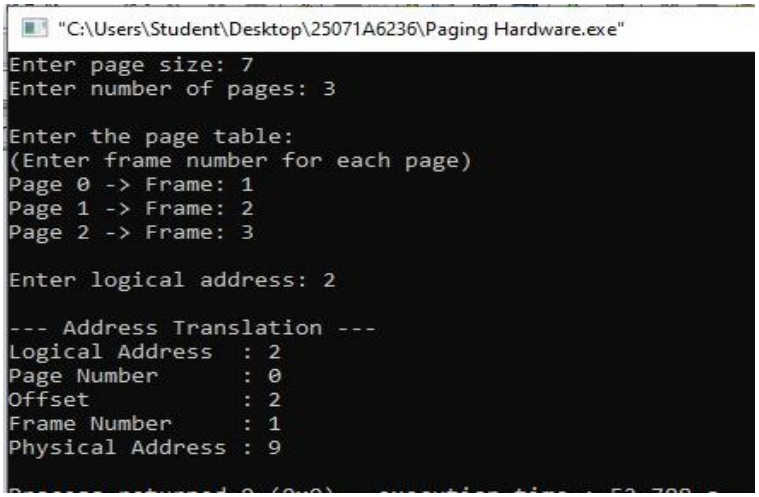
TEST CASE 1:



TEST CASE 2:



TEST CASE 3:



VNR VJIET

Name of the Laboratory:

_____

Name of the Experiment:_____

Experiment No:_____ **Date:**_____

VNR VJIET

Name of the Laboratory:

_____

Name of the Experiment:_____

Experiment No:_____ **Date:**_____